

AI Ethics

Course Num: SEIS 651-01

Instructors: Brandan Keaveny & Jessi Benzel

Week 1

Agenda

Focus Area

- Foundations of AI

This is my favorite week because:

- We're defining the scope of the rest of our semester.
- And I love a good definition.

Week 1

Outline – Key Points

- Introductions 😊
- What is AI?
- What is **not** AI?
- Introductory concepts:
 - NIST AI RMF
 - AI Lifecycle
- Wrap up

Your Background

Zoom polls!

- What is your experience level with **AI in general**?
 - 1 – Beginner, 2 – Intermediate, 3 – Expert
- What is your experience level with **AI Ethics**?
 - 1 – Beginner, 2 – Intermediate, 3 – Expert

Your Instructors

- Meet Brandan and Jessi!

Key Course Expectations

- What is AI Ethics all about? HUMAN IMPACT.
 - We aren't building AI for the sake of AI...
- Effort:
 - 3 hours of study, preparation, etc., per week per credit
 - 3 credit course ~ 9 hours of mostly non-classroom work
 - More critical evaluation, less memorization (and no coding 😞)
- Generative AI:
 - We do not use generative AI to develop this course or for grading
 - We expect that you are not using generative AI to generate your ideas

Lecture: What is AI?

...And what is it NOT?

Background – AI History

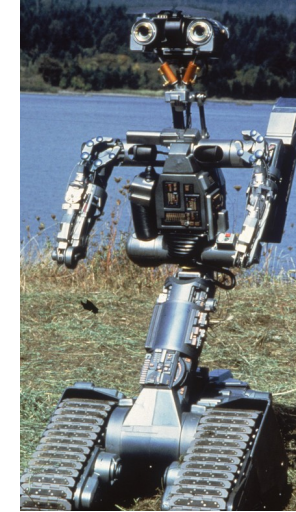
- **Turing Test, 1950:**
 - <https://www.youtube.com/watch?v=3wLqsRLvV-c>
 - <https://courses.cs.umbc.edu/471/papers/turing.pdf>
- **Dartmouth conference, 1956:**
 - <http://jmc.stanford.edu/articles/dartmouth/dartmouth.pdf>
 - “We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to understand how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.”
- **Key Takeaway:** *The concept of AI is not new, but the ability to implement it at scale is.*

AI in Media and Pop Culture



<https://www.linkedin.com/pulse/how-studying-ai-ruins-marvel-movies-you-tony-starks-jarvis-feroli>

[The Jetsons Clip Art | Cartoon Clip Art \(cartoon-clipart.co\)](#)



<https://www.imdb.com/title/tt0091949/>



https://en.wikipedia.org/wiki/Marvin_the_Paranoind_Android



[https://en.wikipedia.org/wiki/Skynet_\(Terminator\)](https://en.wikipedia.org/wiki/Skynet_(Terminator))



https://youtu.be/N43MW7ehkR8?si=-P4X7yTYJDr_i-Rd



<https://www.imdb.com/title/tt2395427/mediaviewer/rm900264448/>



[gryzzl logo parks and rec - Search Images \(bing.com\)](#)

Defining AI

5 U.S.C. 9401(3):

(3) ARTIFICIAL INTELLIGENCE: The term “artificial intelligence” means a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations or decisions influencing real or virtual environments. Artificial intelligence systems use machine and human-based inputs to-

(A) perceive real and virtual environments;

(B) abstract such perceptions into models through analysis in an automated manner; and

(C) use model inference to formulate options for information or action.

Defining AI – Government

- NIST [AI RMF 1.0](#)
 - "...AI systems are designed to operate with varying levels of autonomy..."
- NY State Policy [NYS-P24-001](#)
 - "...The definition includes but is not limited to systems that use machine learning, large language model, natural language processing, and computer vision technologies, including generative AI. The definition does not include basic calculations, basic automation, or pre-recorded “if this then that (IFTT)” response systems."

Defining AI - Industry

- Amazon:
 - Artificial intelligence (AI) is the field of computer science dedicated to solving cognitive problems commonly associated with human intelligence, such as learning, creation, and image recognition. ... The goal with AI is to create self-learning systems that derive meaning from data.
- Microsoft:
 - It's the capability of a computer system to mimic human-like cognitive functions such as learning and problem-solving.
- **"a set of algorithms that try to mimic human intelligence"**
(Dr. Manjeet Rege, UST)

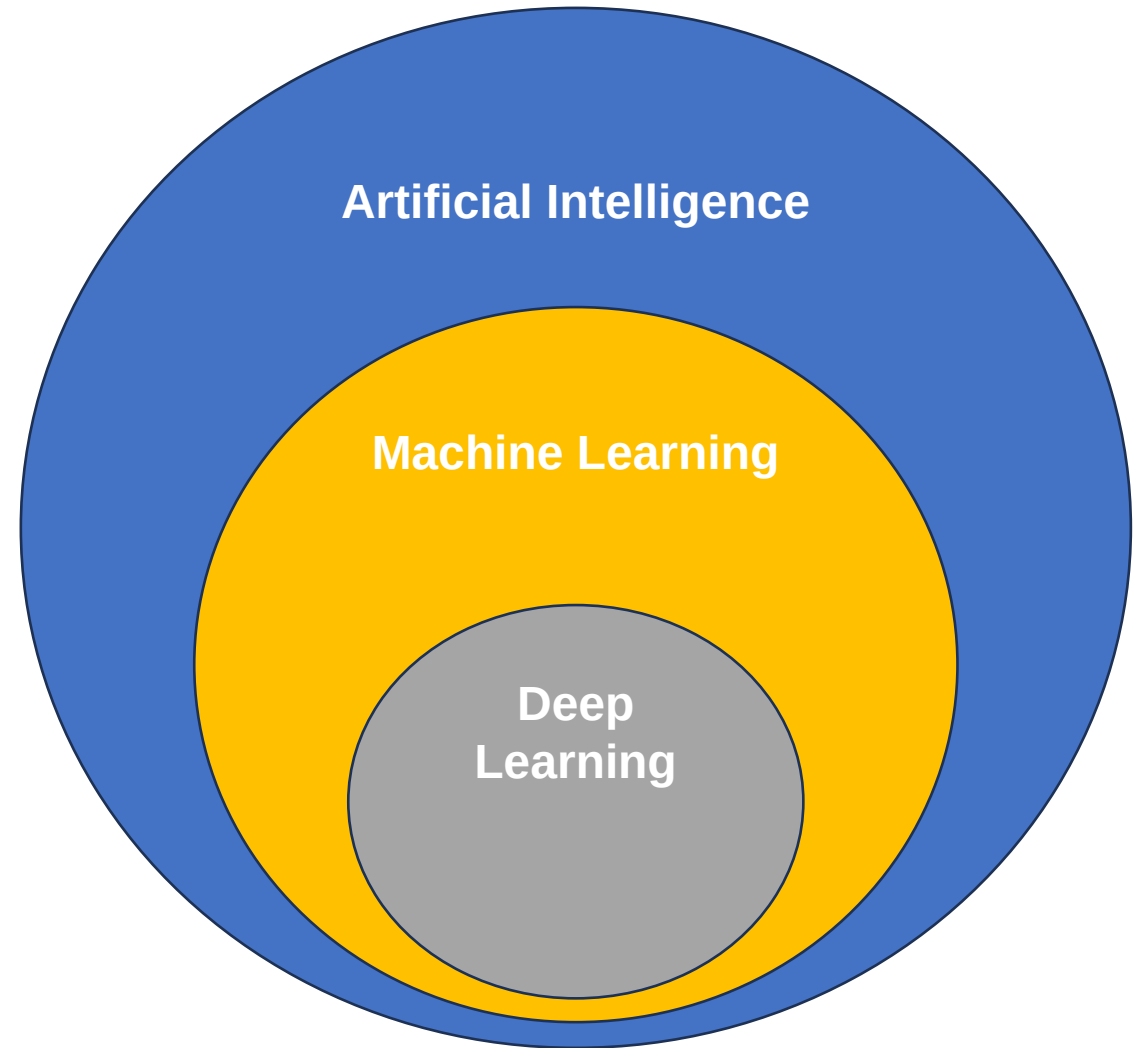
Key Characteristics of AI

- Use of Data (training, learning)
- Excels with "big" and complex data
 - Is the internet "intelligent"?
- Back to the Turing Test
 - Have ChatGPT, Claude, Grok, Gemini, etc., passed the Turing Test?

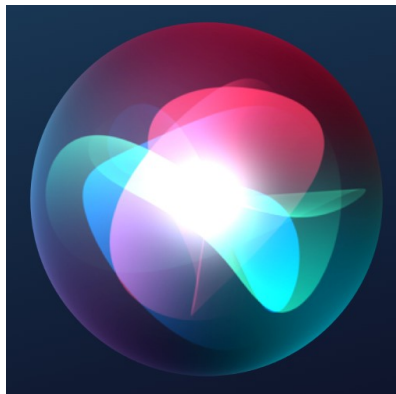
Applications of AI

What about...

- Natural Language Processing
- Image, sound, video recognition
- Robotics
 - Are all robots AI?
- Neural Networks
- "Narrow" vs "General" intelligence
 - Or "Super" intelligence?
- *Where could these fall in our diagram?*
- *Can you have AI without learning?*



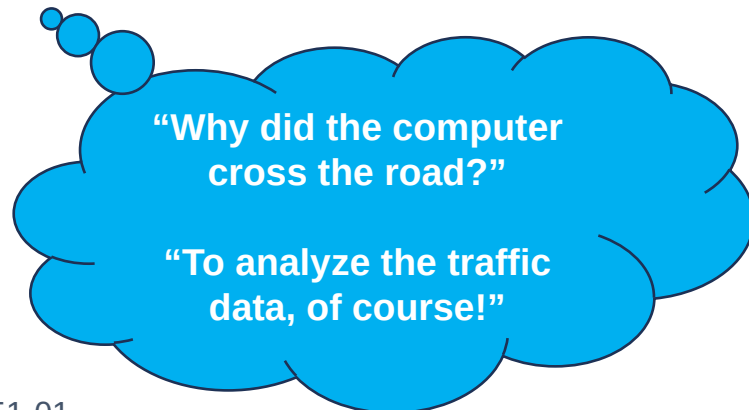
AI in Practice



STITCH FIX

AI in Practice

- Marketing/Advertising
- Recommendation systems
- Loan approvals
- Text to speech
- Generative AI
 - Image, audio, video
 - Language
- GOOGLE MAPS!
- Image recognition
- Autonomous(-ish) vehicles
- Pharmaceutical research
- Predictive/prescriptive analytics
- Even your HR department could be using AI!



What ISN'T AI?

- Isn't AI just “good programming”? A bunch of complicated if statements?
- What about spreadsheets and the sensors in our driers?
 - *Thinking about ethics... where is that data going?*
- It is NOT a magic wand
- AI is not human, it is not alive

Reflect...

- Why didn't NIST (or other governing bodies and industry leaders) define AI in more technical terms
- Why does the definition matter
- What does this have to do with ethics?



AI Lifecycle

- NIST AI Risk Management Framework (Figure 2, pg. 10)
- Test, Evaluate, Verify, Validate (TEVV)
- Note: most of your work is "data" work!
- Don't forget sunseting!



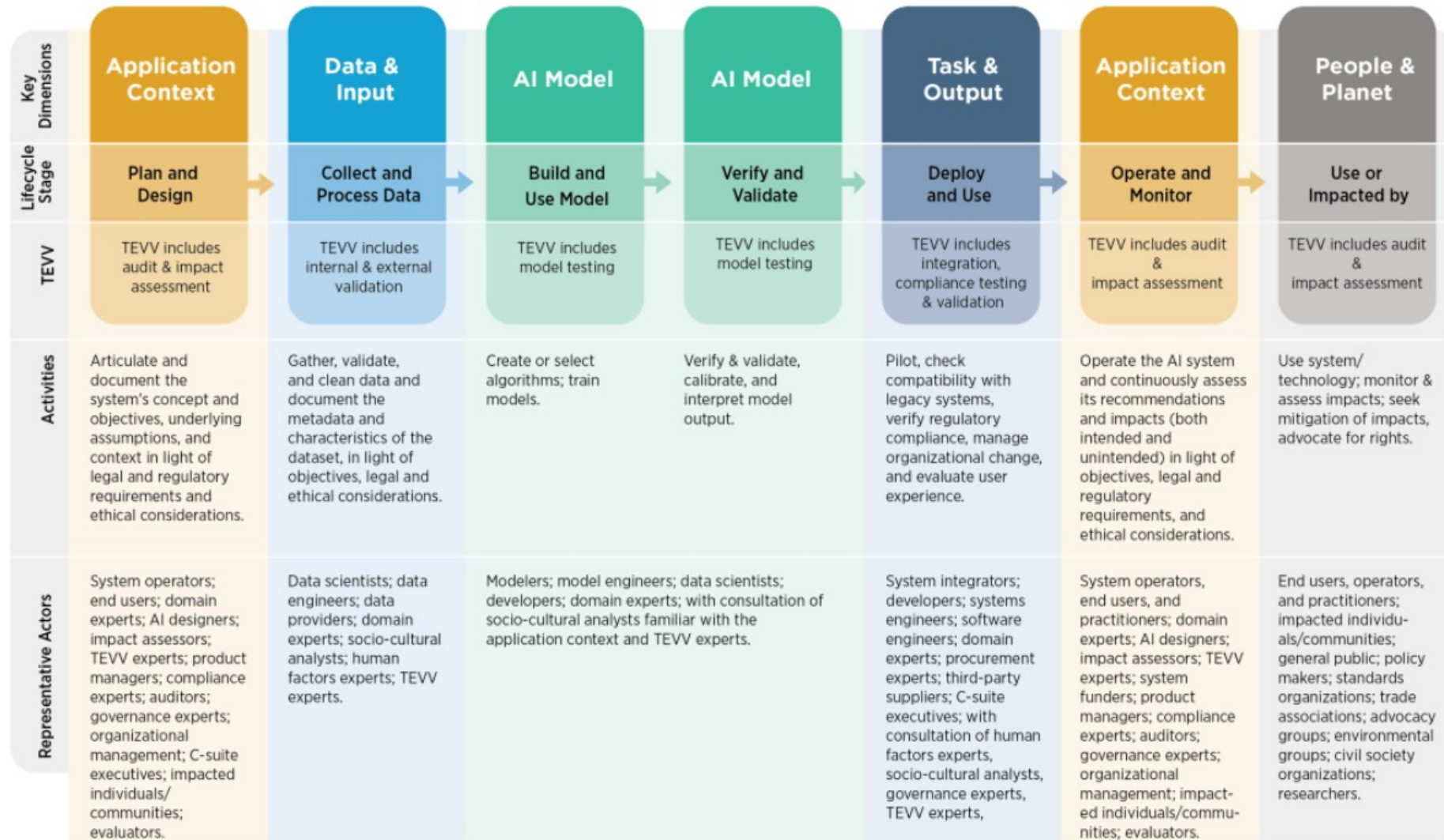
Fig. 2. Lifecycle and Key Dimensions of an AI System. Modified from OECD (2022) [OECD Framework for the Classification of AI systems — OECD Digital Economy Papers](#). The two inner circles show AI systems' key dimensions and the outer circle shows AI lifecycle stages. Ideally, risk management efforts start with the Plan and Design function in the application context and are performed throughout the AI system lifecycle. See Figure 3 for representative AI actors.

[Artificial Intelligence Risk Management Framework \(AI RMF 1.0\) \(nist.gov\)](#)

AI Lifecycle

NIST AI Risk Management Framework (Figure 3, pg. 11)

[Artificial Intelligence Risk Management Framework \(AI RMF 1.0\) \(nist.gov\)](https://nist.gov/artificial-intelligence-risk-management-framework-ai-rmf-1.0)



Challenges in Practice

- "Everybody wants to do the model work, but nobody wants to do the data work"
- "Move fast and break things"
 - "When does the human cost become too high for the building of a better machine?"
 - Minimum Viable vs. Minimum Virtuous?
 - Balance: How not to be a wet blanket
- Shiny New Toys
 - Is AI Ethics a shiny new toy?
- A lot of hype without a lot of knowledge. Do we even know what we're asking for?



Recap - Week 1

- AI is a broad field with varied applications
- Recognizing ethical vs. unethical practices across the AI lifecycle
- **Human impact!**
 - If we're going to do this, let's do it right!

AI work is
***"too important to
be done poorly!"***

References

- [NIST Artificial Intelligence Risk Management Framework \(AI RMF 1.0\) \(nist.gov\)](https://nist.gov/artificial-intelligence/risk-management-framework)
- [nys-p24-001-acceptable-use-of-artificial-intelligence-technologies-_1.pdf](#)
- [What is Artificial Intelligence \(AI\)? - Artificial Intelligence Explained - AWS \(amazon.com\)](#)
- [What is Artificial Intelligence? | Microsoft Azure](#)
- [Data Science in Context](#)
 - Alfred Z. Spector, Massachusetts Institute of Technology, Peter Norvig, Stanford University, California, Chris Wiggins, Columbia University, New York, Jeannette M. Wing, Columbia University, New York

Optional Further Reading:

- [The Era of “Move Fast and Break Things” Is Over \(hbr.org\)](#)
- [A smoother ride and a more detailed Map thanks to AI \(blog.google\)](#)

Class Activity

Skynet

Are We At Skynet?

- What is Skynet
 - <https://www.youtube.com/watch?v=4DQsG3TKQ0I>
- Apply tonight's material: Is Skynet AI?
- Is AI Skynet?
- **Why are we talking about Skynet?**
 - Framing these tools as they really are (versus a fictional or sci-fi character) serves multiple purposes:
 - Reminds us what our role is and where they belong in our society. Namely: PEOPLE are in the driver's seat.
 - Helps us put AI Ethics issues into perspective, and distinguish fiction and hype from reality.